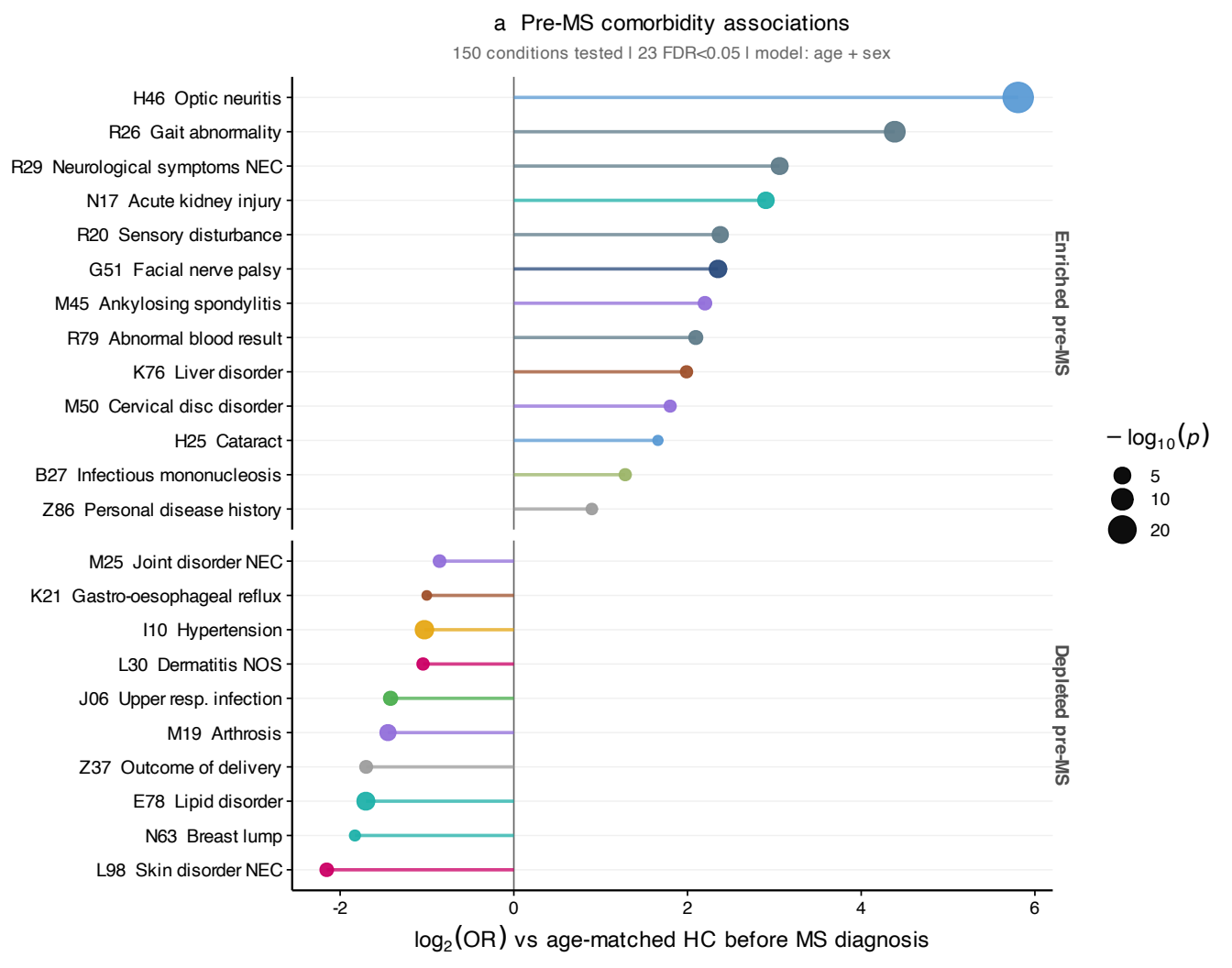
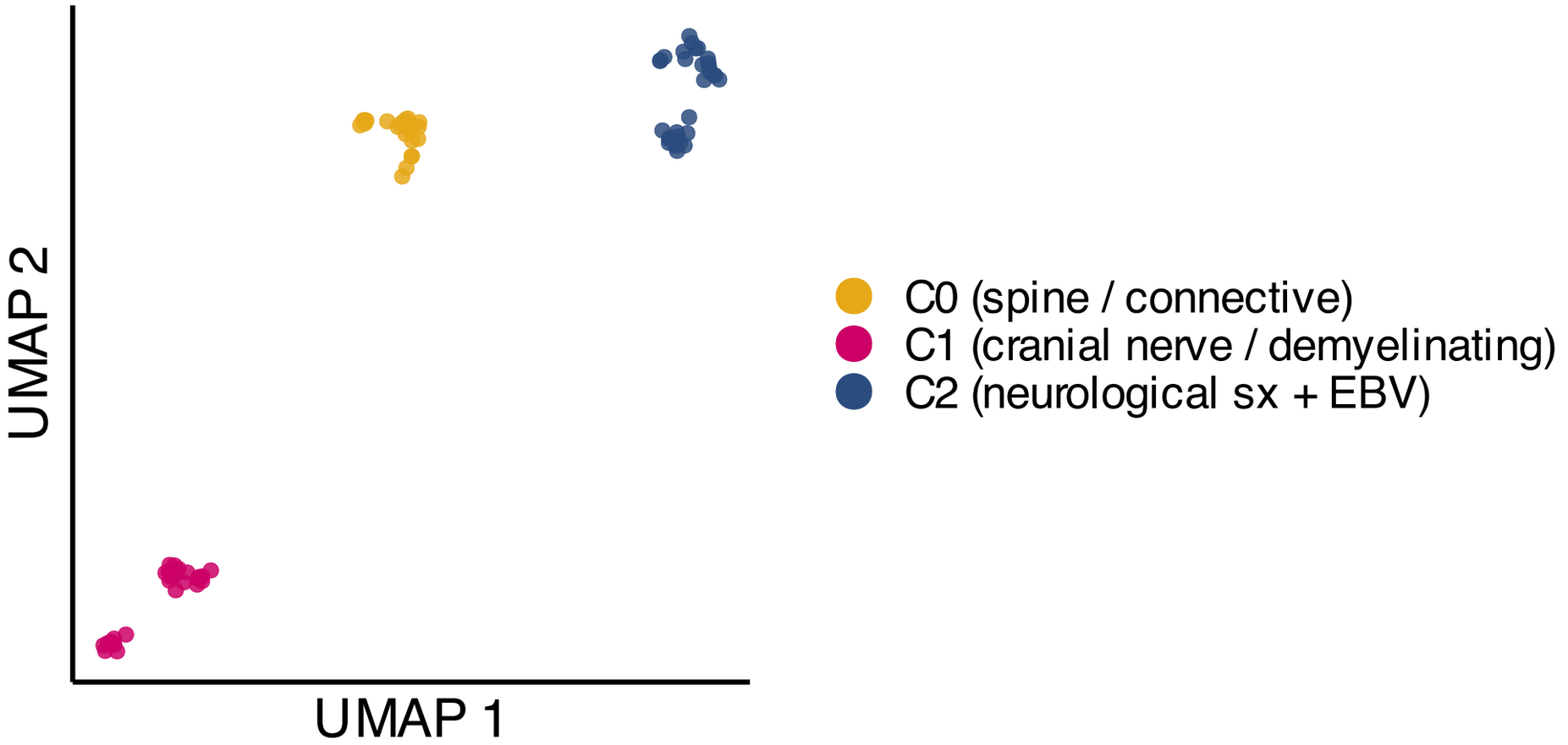
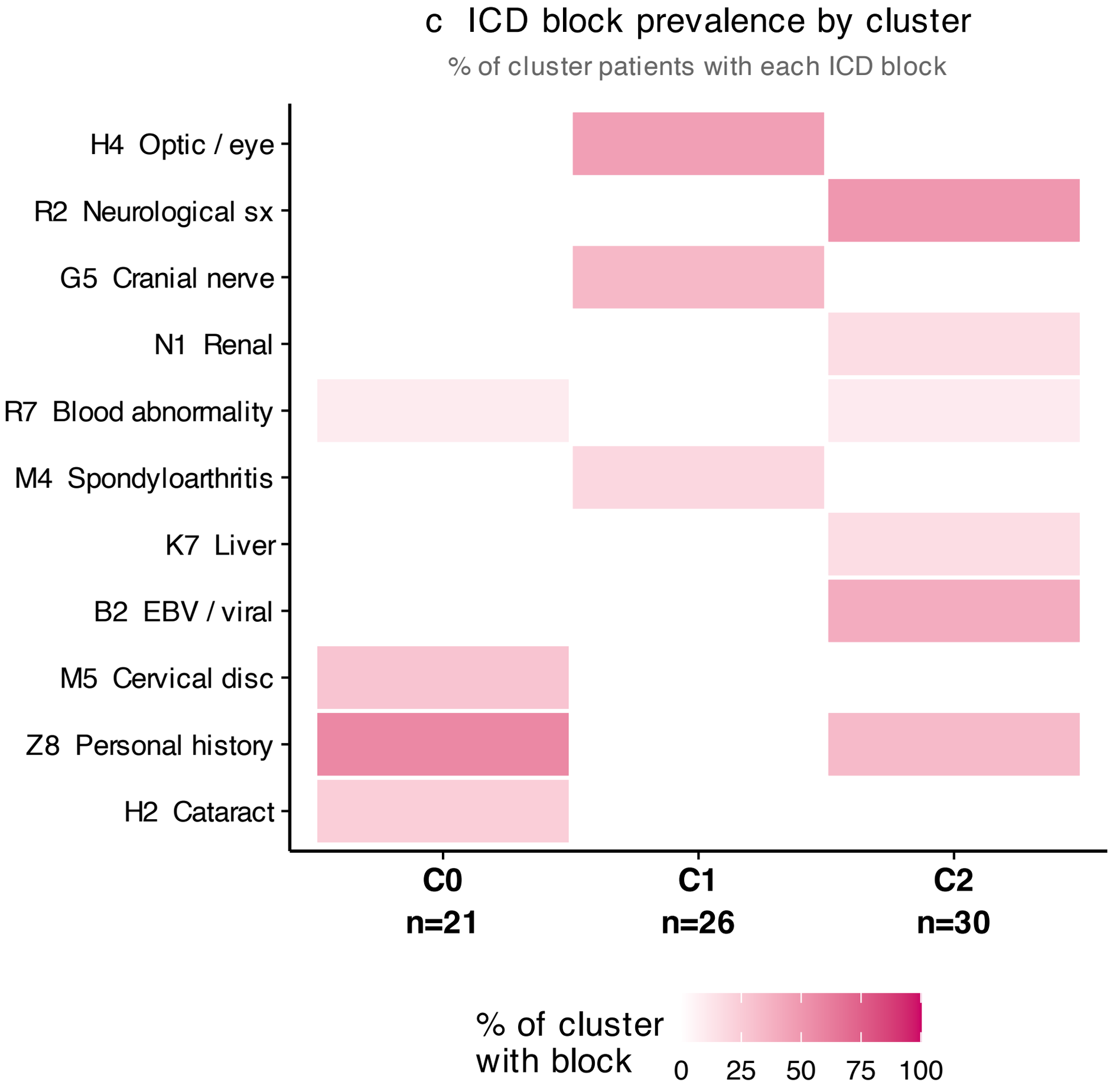
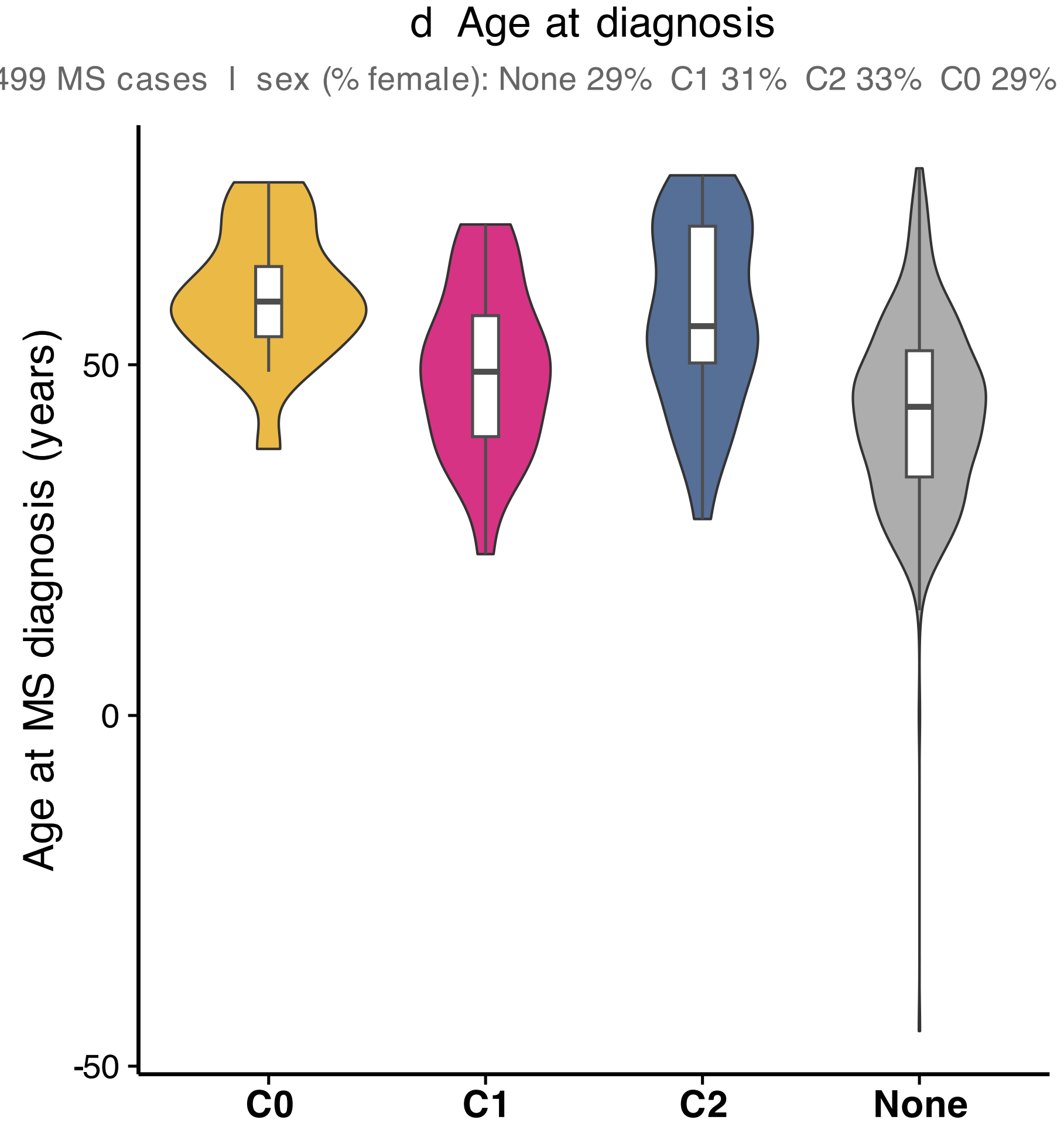

a Pre-MS comorbidity associations
150 conditions tested | 23 FDR<0.05 | model: age + sex
H46 Optic neuritis
R26 Gait abnormality
R29 Neurological symptoms NEC
N17 Acute kidney injury
R20 Sensory disturbance
G51 Facial nerve palsy
M45 Ankylosing spondylitis
R79 Abnormal blood result
K76 Liver disorder
M50 Cervical disc disorder
H25 Cataract
B27 Infectious mononucleosis
Z86 Personal disease history
M25 Joint disorder NEC
K21 Gastro-oesophageal reflux
I10 Hypertension
L30 Dermatitis NOS
J06 Upper resp. infection
M19 Arthrosis
Z37 Outcome of delivery
E78 Lipid disorder
N63 Breast lump
L98 Skin disorder NEC
Enriched pre-MS
Depleted pre-MS
-2
0
2
4
6
log2(OR) vs age-matched HC before MS diagnosis
−log10(p)
5
10
20

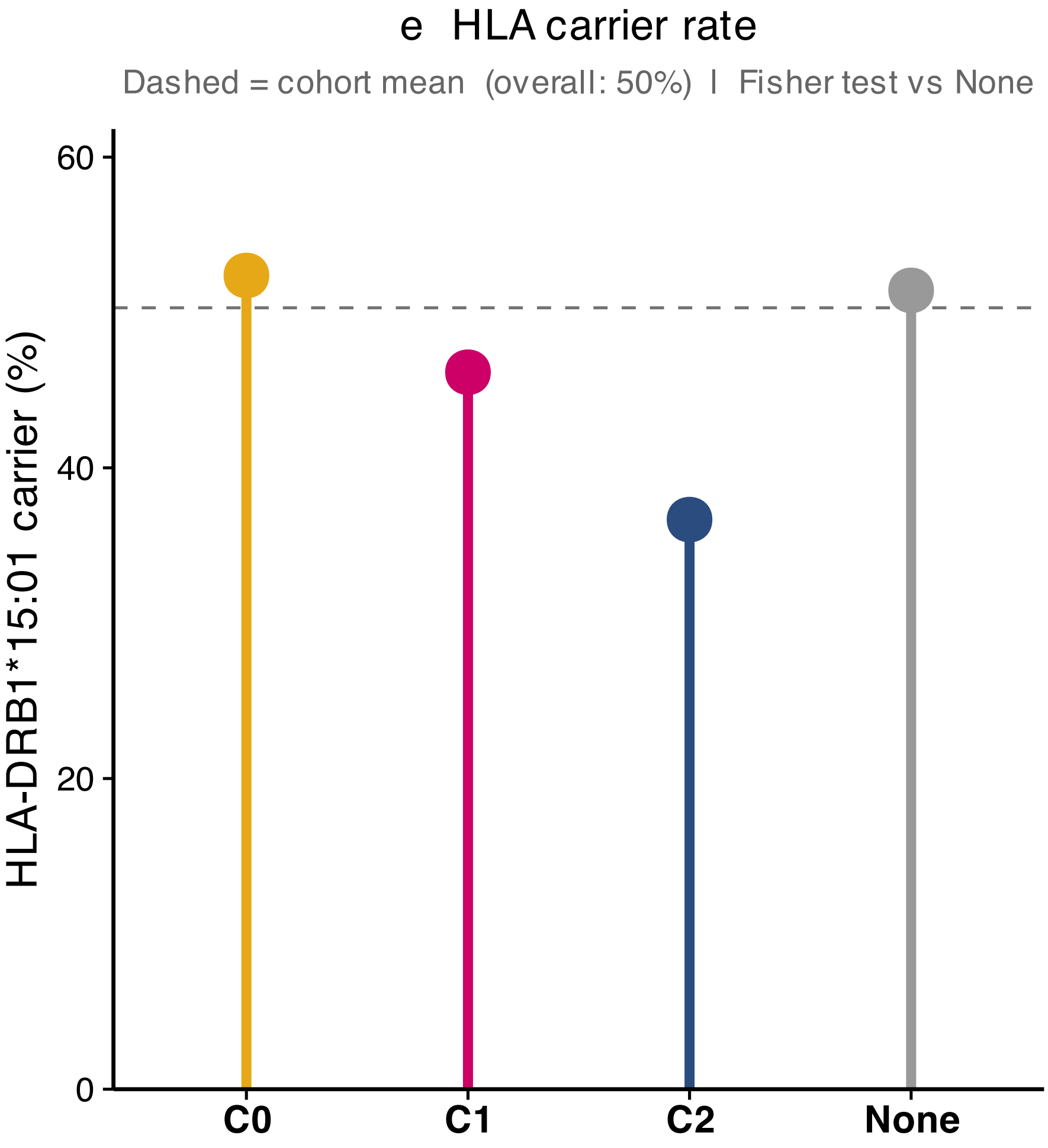
b Pre-MS endophenotype clusters

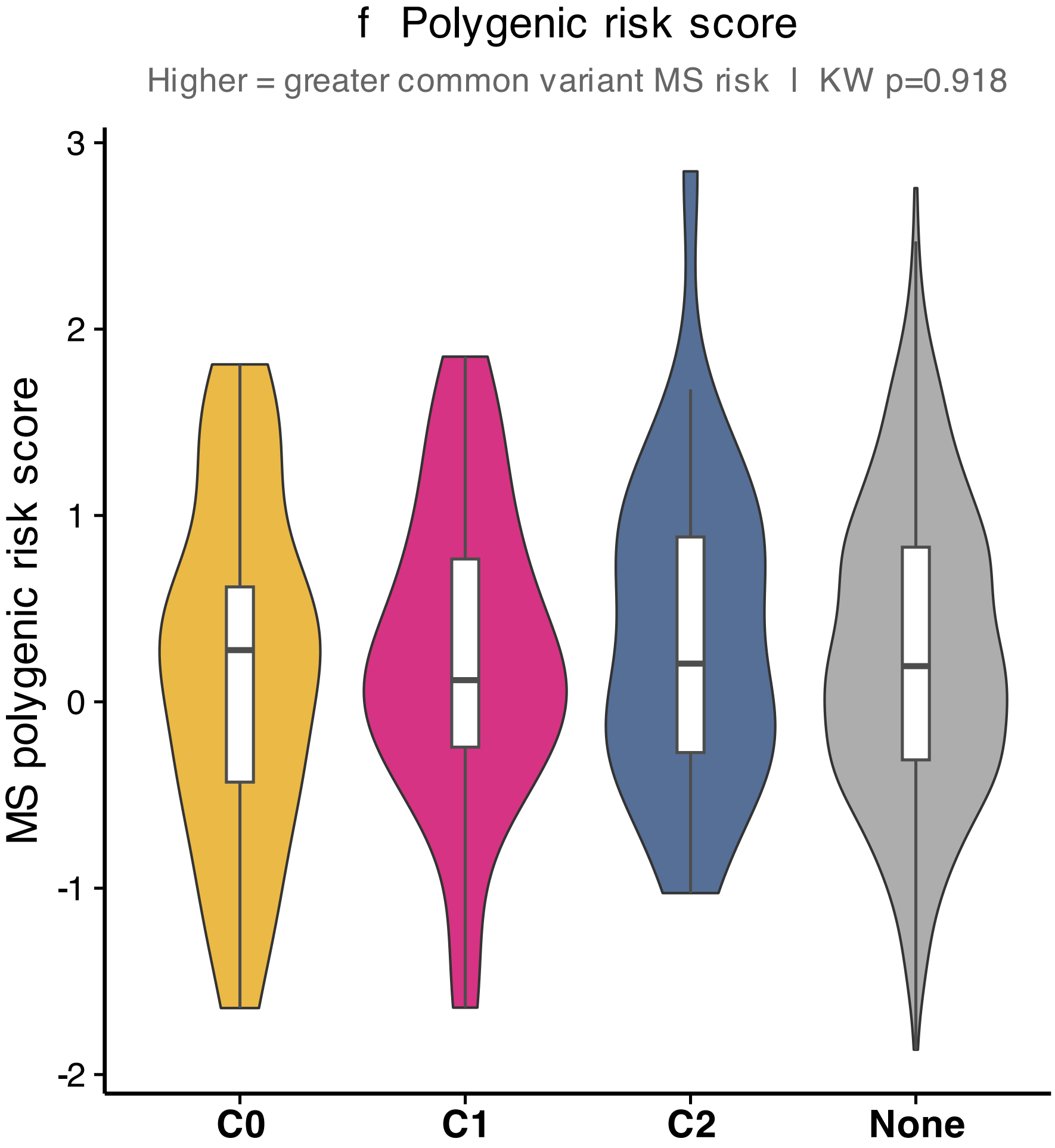
l=0.86 l n=77 patients with pre-MS features (n=422 without not shown)

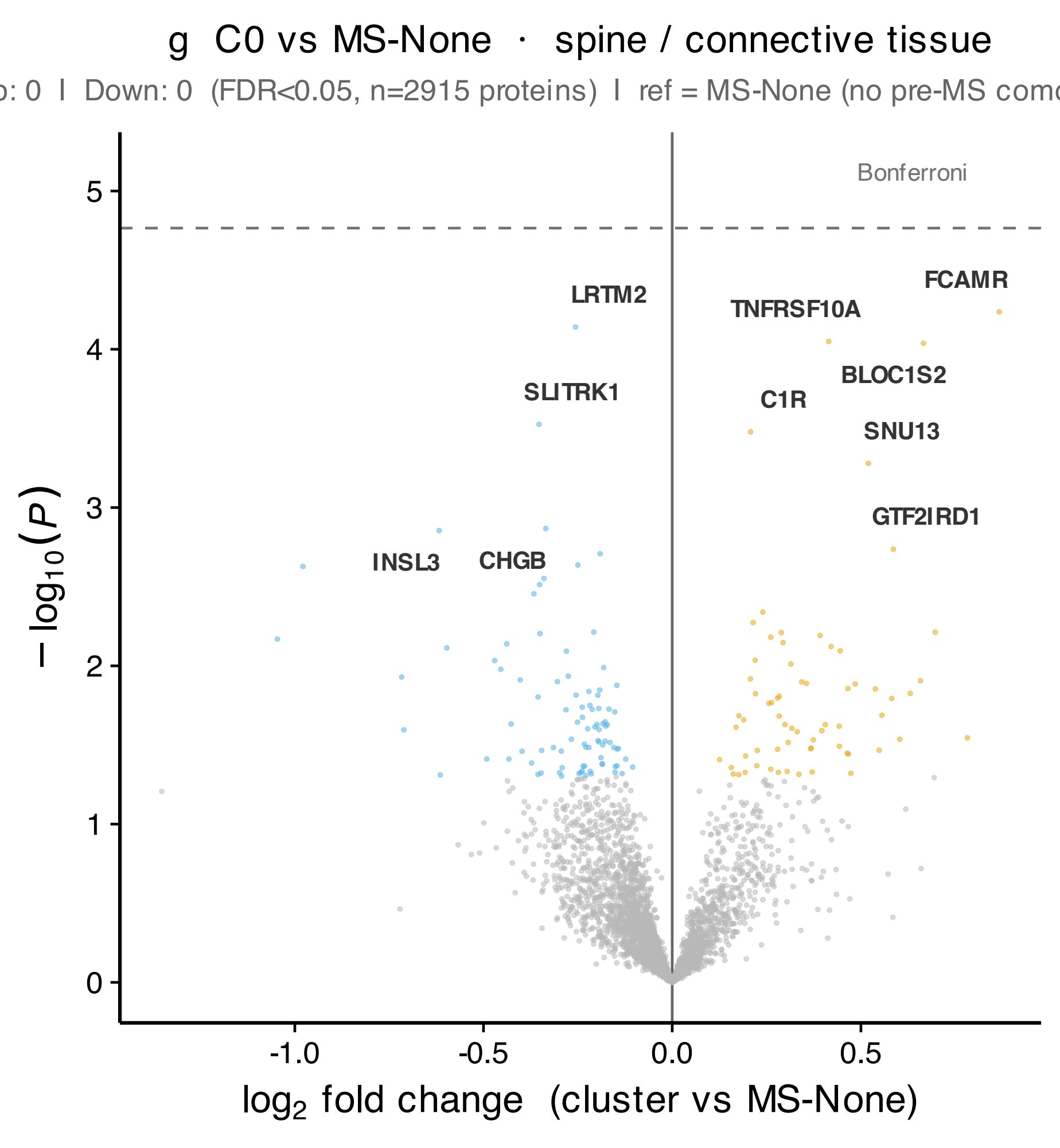


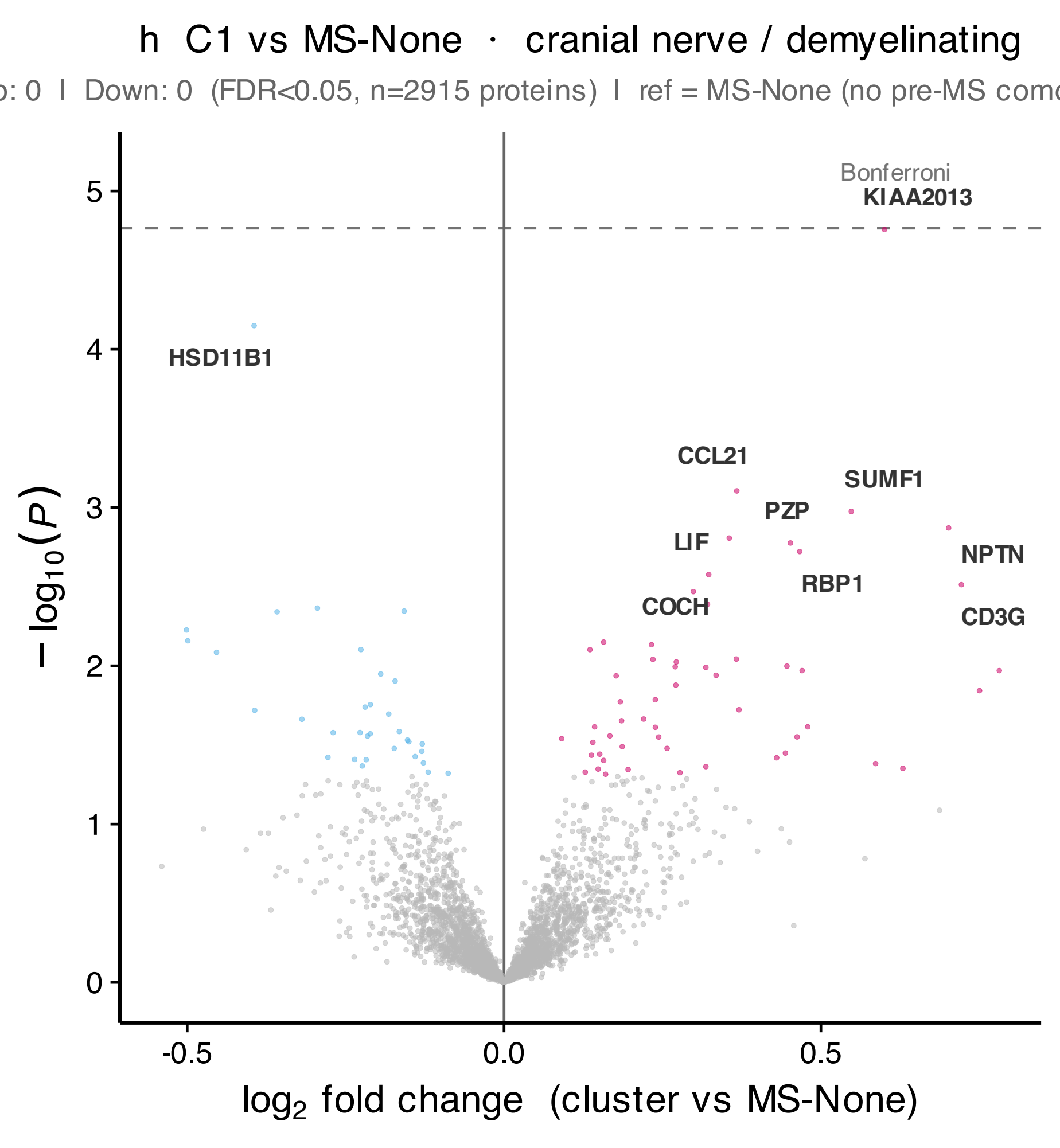

c ICD block prevalence by cluster
% of cluster patients with each ICD block
H4 Optic / eye
R2 Neurological sx
G5 Cranial nerve
N1 Renal
R7 Blood abnormality
M4 Spondyloarthritis
K7 Liver
B2 EBV / viral
M5 Cervical disc
Z8 Personal history
H2 Cataract
C0
n=21
C1
n=26
C2
n=30
% of cluster with block
0
25
50
75
100

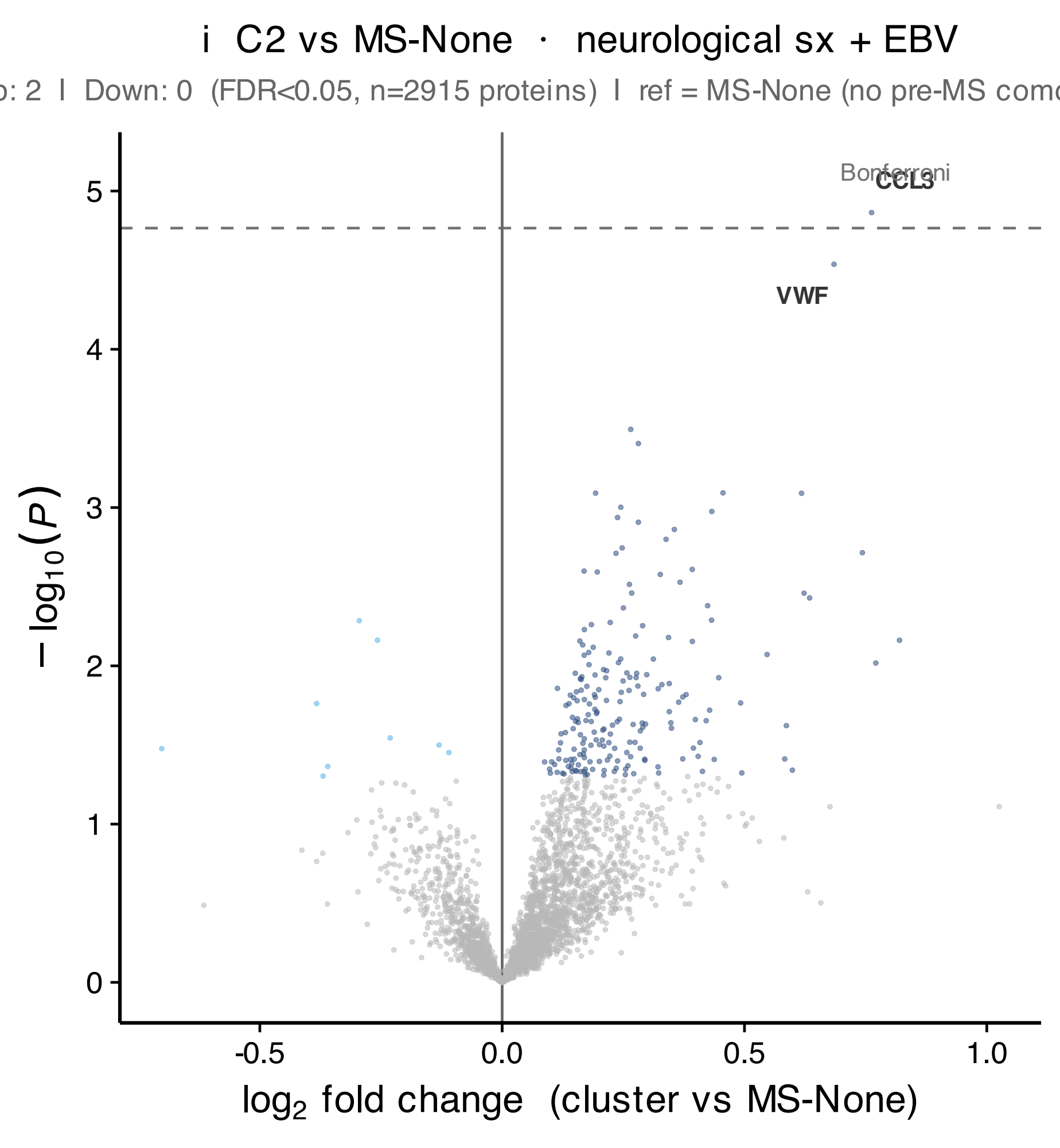

d Age at diagnosis
499 MS cases | sex (% female): None 29% C1 31% C2 33% C0 29%
Age at MS diagnosis (years)
50
0
-50
C0
C1
C2
None

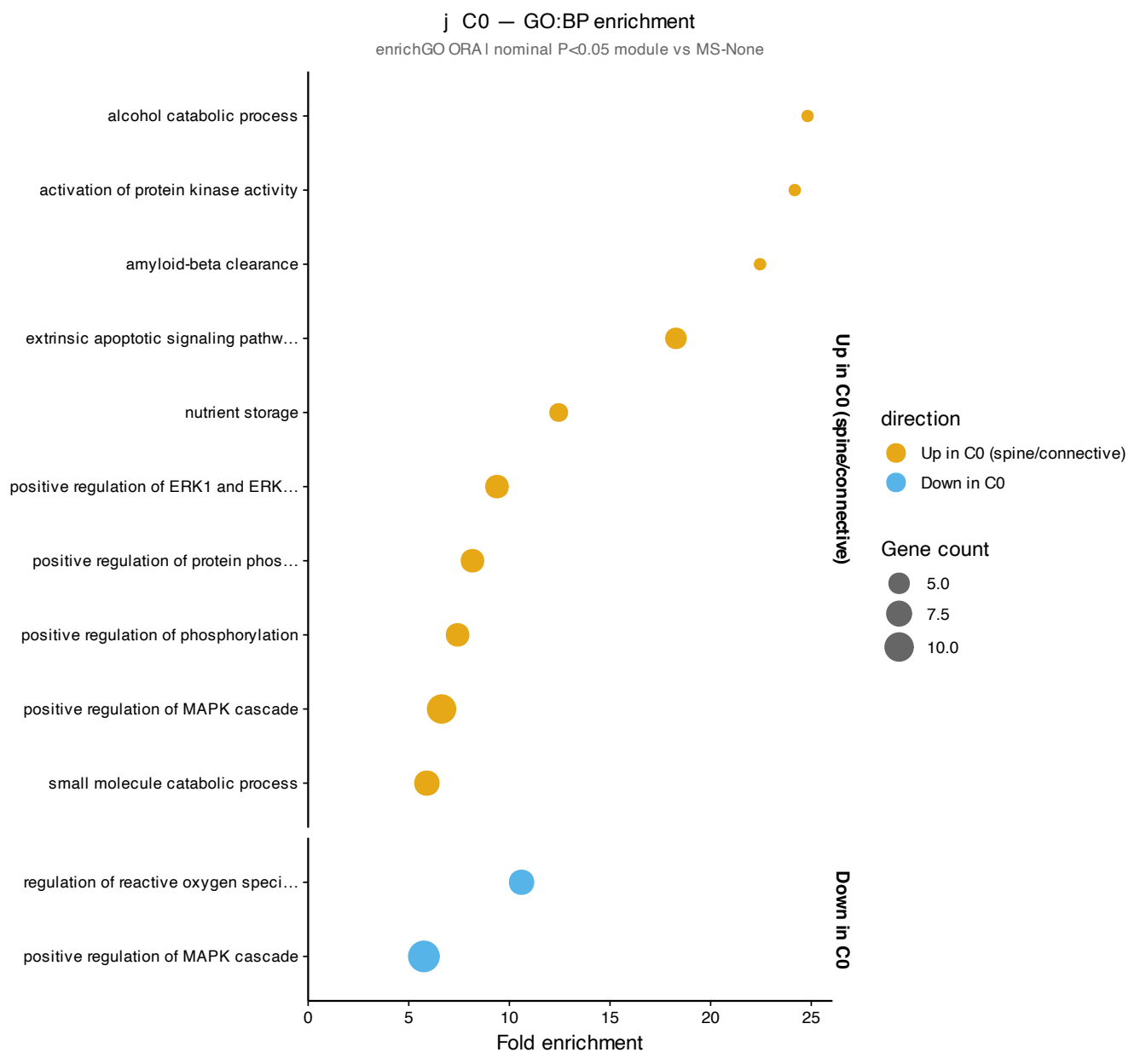

e HLA carrier rate
Dashed = cohort mean (overall: 50%) | Fisher test vs None
HLA-DRB1*15:01 carrier (%)
60
40
20
0
C0
C1
C2
None

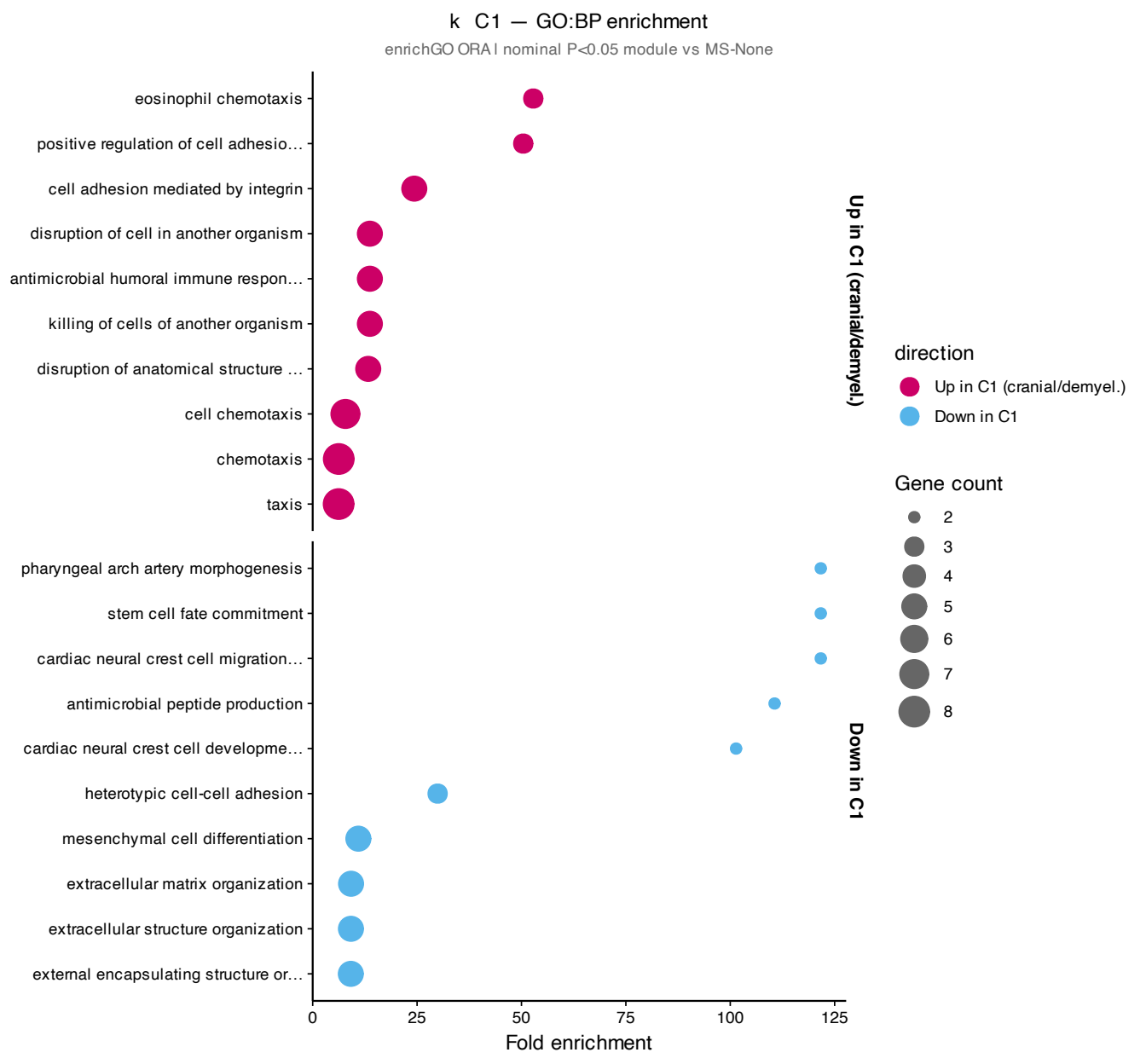

f Polygenic risk score
Higher = greater common variant MS risk | KW p=0.918
MS polygenic risk score
3
2
1
0
-1
-2
C0
C1
C2
None

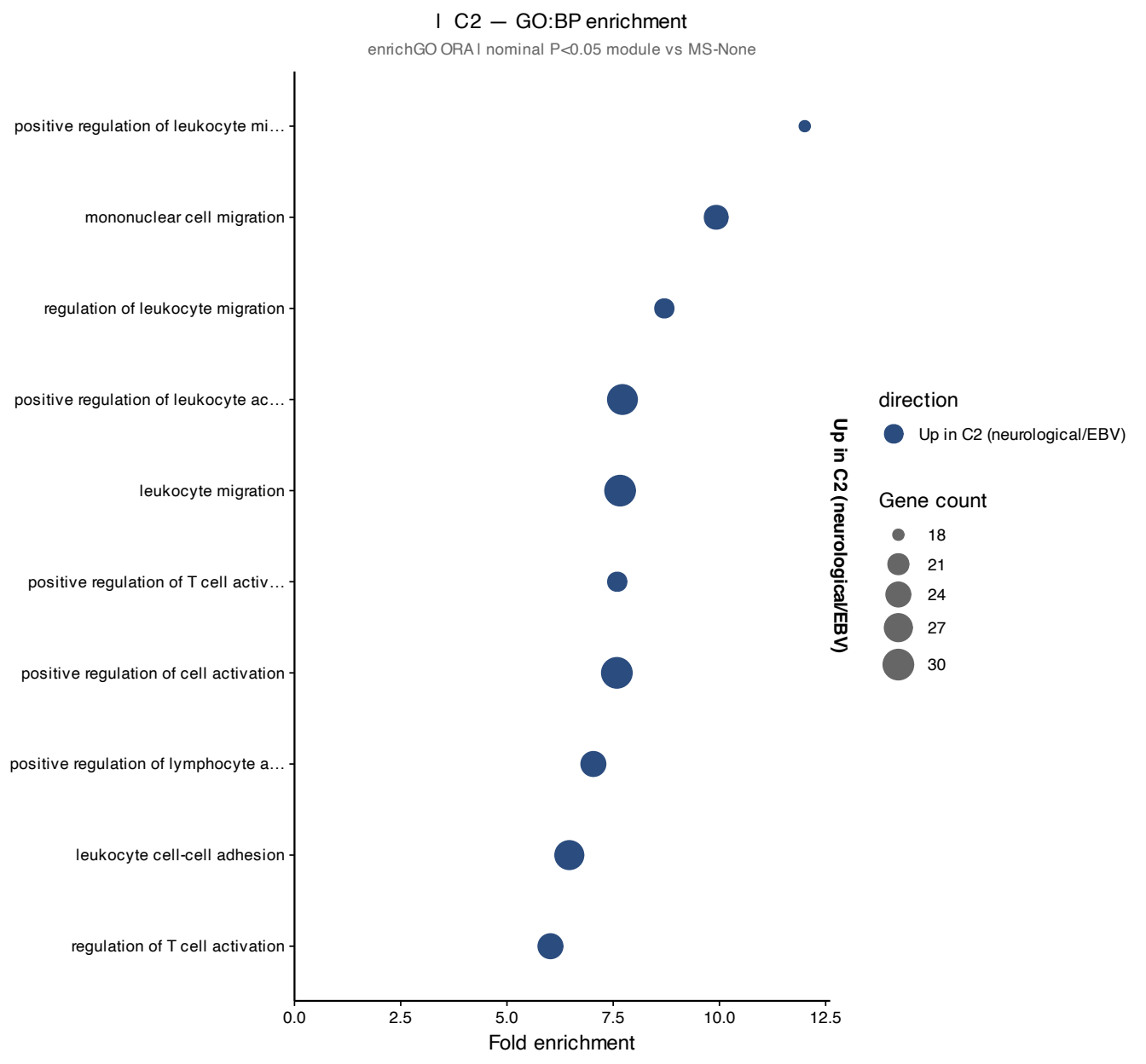

g C0 vs MS-None · spine / connective tissue
p: 0 | Down: 0 (FDR<0.05, n=2915 proteins) | ref = MS-None (no pre-MS como
Bonferroni
FCAMR
LRTM2
TNFRSF10A
BLOC1S2
SLITRK1
C1R
SNU13
GTF2IRD1
INSL3
CHGB
5
4
3
2
1
0
−log10(P)
-1.0
-0.5
0.0
0.5
log2 fold change (cluster vs MS-None)


h C1 vs MS-None · cranial nerve / demyelinating
p: 0 | Down: 0 (FDR<0.05, n=2915 proteins) | ref = MS-None (no pre-MS como
Bonferroni
KIAA2013
HSD11B1
CCL21
SUMF1
PZP
LIF
NPTN
RBP1
COCH
CD3G
5
4
3
2
1
0
−log10(P)
-0.5
0.0
0.5
log2 fold change (cluster vs MS-None)


i C2 vs MS-None · neurological sx + EBV
p: 2 | Down: 0 (FDR<0.05, n=2915 proteins) | ref = MS-None (no pre-MS como
Bonferroni
CCL3
VWF
5
4
3
2
1
0
−log10(P)
-0.5
0.0
0.5
1.0
log2 fold change (cluster vs MS-None)


j C0 — GO:BP enrichment
enrichGO ORA | nominal P<0.05 module vs MS-None
alcohol catabolic process
activation of protein kinase activity
amyloid-beta clearance
extrinsic apoptotic signaling pathw…
nutrient storage
positive regulation of ERK1 and ERK…
positive regulation of protein phos…
positive regulation of phosphorylation
positive regulation of MAPK cascade
small molecule catabolic process
regulation of reactive oxygen speci…
positive regulation of MAPK cascade
Up in C0 (spine/connective)
Down in C0
0
5
10
15
20
25
Fold enrichment
direction
Up in C0 (spine/connective)
Down in C0
Gene count
5.0
7.5
10.0


k C1 — GO:BP enrichment
enrichGO ORA | nominal P<0.05 module vs MS-None
eosinophil chemotaxis
positive regulation of cell adhesio...
cell adhesion mediated by integrin
disruption of cell in another organism
antimicrobial humoral immune respon...
killing of cells of another organism
disruption of anatomical structure ...
cell chemotaxis
chemotaxis
taxis
pharyngeal arch artery morphogenesis
stem cell fate commitment
cardiac neural crest cell migration...
antimicrobial peptide production
cardiac neural crest cell developme...
heterotypic cell-cell adhesion
mesenchymal cell differentiation
extracellular matrix organization
extracellular structure organization
external encapsulating structure or...
Up in C1 (cranial/demyel.)
Down in C1
0
25
50
75
100
125
Fold enrichment
direction
Up in C1 (cranial/demyel.)
Down in C1
Gene count
2
3
4
5
6
7
8


I C2 — GO:BP enrichment
enrichGO ORA | nominal P<0.05 module vs MS-None
positive regulation of leukocyte mi…
mononuclear cell migration
regulation of leukocyte migration
positive regulation of leukocyte ac…
leukocyte migration
positive regulation of T cell activ…
positive regulation of cell activation
positive regulation of lymphocyte a…
leukocyte cell-cell adhesion
regulation of T cell activation
0.0
2.5
5.0
7.5
10.0
12.5
Fold enrichment
Up in C2 (neurological/EBV)
direction
Up in C2 (neurological/EBV)
Gene count
18
21
24
27
30